Contents

1 Introduction to Groups

1.1 Groups

A group is a higher abstraction for a structure which supports a certain arithmetic operation. To fully specify a group, we need to abstract the notion of an "arithmetic operation".

Definition 1.1.1 ▸ Binary Operation

A **binary operation** $\star$ on a set G is a function $f : G \times G \to G$.

Conventionally, the notations $a \star b$ and $\star\big((a, b)\big)$ are equivalent. For the sake of simplicity, we will not use the latter.

Recall that there are some commonly used properties of operations. We shall abstract those as well. First, we pay attention to whether the order of execution matters in a sequence of repeated applications of an operation.

Definition 1.1.2 ▸ Associativity

A binary operation $\star$ on G is **associative** if for all $a, b, c \in G$, we have

$$a \star (b \star c) = (a \star b) \star c.$$

Similarly, we will next focus on whether the order of operands matters when an operation is applied.

Definition 1.1.3 ▸ Commutativity

Let G be a set with a binary operation $\star$, then $a, b \in G$ **commute** if $a \star b = b \star a$. We say that $\star$ or G is **commutative** if all elements in G commute pair-wisely with respect to $\star$.

Note that both addition and multiplication are associative and commutative over $\mathbb{N}$, but subtraction is not. In fact, subtraction is not even a binary operation over $\mathbb{N}$ because $\mathbb{N}$ is not closed under subtraction.

A more complex example is cross product in $\mathbb{R}^n$, which is neither associative nor commutative. Recall that we used the word "closed" when arguing that subtraction is not a binary operation over $\mathbb{N}$. We shall further formalise this notion.

Definition 1.1.4 ▶ Closure

Let $\star$ be a binary operation on G, then $H \subseteq G$ is **closed under $\star$** if for all $a, b \in H$, we have $a \star b \in H$.

In particular, this means that $\star$ on G is still a binary operation on any $H \subseteq G$ which is closed under $\star$. Intuitively, both associativity and commutativity can be inherited by these subsets.

Proposition 1.1.5 ▶ Presevation of Associativity and Commutativity

Let $\star$ be a binary operation on G and $H \subseteq G$ be closed under $\star$, then

- *if $\star$ is associative on G, then $\star$ is associative on H;*
- *if $\star$ is commutative on G, then $\star$ is commutative on H.*

Proof. Left as exercise to the reader. □

Next, we will introduce the fundamental axioms which define the abstract mathematical notion of groups, as well as some simple results derived from these axioms.

Definition 1.1.6 ▶ Group

A **group** is a pair $(G, \star)$ where G is a set and $\star$ is a binary operation on G such that

- $\star$ is associative;
- there exists some $e \in G$ with $a \star e = e \star a = a$ for all $a \in G$, where e is known as an **identity** of G;
- for all $a \in G$, there exists some $a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$, where a^{-1} is known as the **inverse** of a.

In Definition 1.1.6, the map $\star$ is often called *multiplication* by convention. Moreover, we can define a map $^{-1} : G \to G$ known as the *inversion* map as an alternative way to construct the inverses.

Therefore, a group can be defined as a set in which:

1. any two elements can be multiplied,
2. every element has an inverse in the set itself, and
3. there exists an identity element which equals the product of any element multiplied by its inverse and multiplied by which any element equals itself.

The very first thing to take note here is that the order of multiplication matters in groups! In other words, $a \star b \neq b \star a$ in general.

Definition 1.1.7 ▸ Abelian Group

A group G is called an **Abelian** or **commutative** group if it is commutative.

Another important thing to remember here is that, although we call the map $\star$ "multiplication" conventionally, it does not have to be referring to the multiplication between real numbers or vectors or anything that we commonly take to be able to be multiplied together in elementary mathematics.

In fact, we will see that by taking $\mathbb{Z}$ as the set and addition $+$ as the multiplication map, we get a group that satisfies Definition 1.1.6 perfectly! Furthermore, this group $(\mathbb{Z}, +)$, known as the *additive group* of $\mathbb{Z}$, is an Abelian group. The same holds for $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$.

Consider $(\mathbb{Q} \setminus \{0\}, +)$, which is not a group due to the missing identity. However, we can turn it into a group, specifically into a *multiplicative group*, by changing the operation to $\times$. In fact, $(\mathbb{Q} \setminus \{0\}, \times)$ is also Abelian. The same holds for $(\mathbb{R} \setminus \{0\}, \times)$ and $(\mathbb{C} \setminus \{0\}, \times)$.

> *Remark.* Writing $\frac{b}{a}$ is not a good notation in the context of this course because it could mean either $a^{-1} \star b$ or $b \star a^{-1}$, which are completely different for non-Abelian groups.

Note that further $(\mathbb{Z} \setminus \{0\}, \times)$ is not a group because all elements except 1 do not have inverses. Therefore, we define $\mathbb{Z}^\times := \{-1, 1\}$ which under $\times$ is a multiplicative group.

A group can be finite or infinite. To define those we need to define a measurement for the "size" of a group.

Definition 1.1.8 ▸ Order of a Group

Let $(G, \star)$ be a group, then $|G|$ is known as the *order* of the group. G is said to be a **finite** group if G is finite, and **infinite** group otherwise.

Let us consider some extreme cases here. Note that a group cannot be empty, so the most extreme case is a singleton group.

Proposition 1.1.9 ▸ Singleton Sets Are Abelian Groups

Every singleton set $\{e\}$ is an Abelian group under any binary operation. In particular, the only binary operation under which the set is a group is $(e, e) \mapsto e$.

Proof. Trivial. □

Next is a more complex example based on the *roots of unity*.

Definition 1.1.10 ▸ Root of Unity

An **n-th root of unity** is a complex number $z \in \mathbb{C}$ such that $z^n = 1$.

We can collect all such roots and realise that they form a group.

Proposition 1.1.11 ▸ n-th Root of Unity Group

Let $\mathbb{Z}_n := \{0, 2, \cdots, n-1\}$ *and define*

$$U_n := \left\{\exp\left(\frac{2k\pi \mathrm{i}}{n}\right) : k \in \mathbb{Z}_n\right\},$$

then $(U_n, \cdot)$ *is a group.*

Proof. We first prove that U_n is closed under multiplication. Take any $z_1, z_2 \in U_n$, then there exist $k_1, k_2 \in \mathbb{Z}_n$ such that $z_1 = \exp\left(\frac{2k_1\pi \mathrm{i}}{n}\right)$ and $z_2 = \exp\left(\frac{2k_2\pi \mathrm{i}}{n}\right)$. Notice that there exists some $q \in \mathbb{N}$ and $r \in \mathbb{Z}_n$ such that $k_1 + k_2 = nq + r$, so

$$\begin{aligned}
z_1 \cdot z_2 &= \exp\left(\frac{2(k_1 + k_2)\pi \mathrm{i}}{n}\right) \\
&= \exp\left(\frac{2(nq + r)\pi \mathrm{i}}{n}\right) \\
&= \exp\left(2q\pi \mathrm{i} + \frac{2r\pi \mathrm{i}}{n}\right) \\
&= \exp(2q\pi \mathrm{i}) \exp\left(\frac{2r\pi \mathrm{i}}{n}\right).
\end{aligned}$$

Observe that $\exp(2q\pi \mathrm{i}) = 1$, so $z_1 \cdot z_2 = \exp\left(\frac{2r\pi \mathrm{i}}{n}\right) \in U_n$. It is obvious that multiplication is associative over $\mathbb{C}$ and $1 \in U_n$ is an identity, so it suffices to prove the existence of inverses. Take any $z \in U_n$, then there exists some $k \in \mathbb{Z}_n$ such that $z = \exp\left(\frac{2k\pi \mathrm{i}}{n}\right)$. Consider

$$\begin{aligned}
z' &:= \exp\left(\frac{-2k\pi \mathrm{i}}{n}\right) \\
&= \exp(2\pi \mathrm{i}) \exp\left(\frac{-2k\pi \mathrm{i}}{n}\right) \\
&= \exp\left(\frac{2(n-k)\pi \mathrm{i}}{n}\right).
\end{aligned}$$

Note that $n - k \in \mathbb{Z}_n$ and $z' \cdot z = z \cdot z' = 1$, so $z' = z^{-1}$. Therefore, $(U_n, \cdot)$ is a group. $\square$

We shall prove some trivial results directly from the definition of groups.

Theorem 1.1.12 ▸ Uniqueness of Identity and Inverse in Groups

Let G be a group under $\star$ with identity e, then e is unique and for all $a \in G$, a^{-1} is unique. Furthermore,

- $\left(a^{-1}\right)^{-1} = a$ *for all* $a \in G$*;*
- $(a \star b)^{-1} = b^{-1} \star a^{-1}$ *for all* $a, b \in G$*.*

Proof. Suppose there exists $f \in G$ such that $a \star f = f \star a = a$ for all $a \in G$. Since $e \in G$, we have $e \star f = e$, but e is the identity and $f \in G$, so $f \star e = f$. Therefore,

$$f = f \star e = e \star f = e,$$

which means that e is unique. Suppose that there is some $a \in G$ such that there is some $b \in G$ with $a \star b = b \star a = e$, then

$$\begin{aligned} b &= b \star e \\ &= b \star \left(a \star a^{-1}\right) \\ &= (b \star a) \star a^{-1} \\ &= e \star a^{-1} \\ &= a^{-1}. \end{aligned}$$

Therefore, a^{-1} is unique for all $a \in G$. □

Theorem 1.1.12 shows that if the map $\star$ is such that an identity exists in the set G, then it is unique, which also uniquely determines the inversion map as there cannot be different $a, b \in G$ such that $ac = bc = e$ for some $c \in G$. In other words, **a group is uniquely determined by the multiplication map** given a set G.

Theorem 1.1.13 ▸ Generalised Associative Law

Let $(G, \star)$ be a group, then $a_1 \star a_2 \star \cdots \star a_n$ is well-defined for any $a_1, a_2, \cdots, a_n \in G$ and all $n \in \mathbb{N}^+$ and $n \geq 3$.

Proof. We shall proceed with induction on n. The case where $n = 3$ is immediate from Definition 1.1.6. Suppose that there exists some $k \in \mathbb{N}^+$ with $k \geq 3$ such that

$$a_1 \star a_2 \star \cdots \star a_k$$

is well-defined for any $a_1, a_2, \cdots, a_k \in G$. Take any $a_1, a_2, \cdots, a_{k+1} \in G$ and write

$$a_1 \star a_2 \star \cdots \star a_k = K \in G,$$

then clearly $K \star a_{k+1} \in G$ and so $a_1 \star a_2 \star \cdots \star a_{k+1}$ is well-defined. $\square$

The above results justify some short-hand notations in a group G.

- We denote the identity of any group by 1.
- If the binary operation is clear, we can omit it by writing $a \star b = a \cdot b = ab$.
- For all $a \in G$ and any $n \in \mathbb{Z}$,

$$a^n = \begin{cases} \underbrace{a \cdot a \cdot \cdots \cdot a}_{n \text{ terms}} & \text{if } n > 0 \\ 1 & \text{if } n = 0 \\ \underbrace{a^{-1} \cdot a^{-1} \cdot \cdots \cdot a^{-1}}_{n \text{ terms}} & \text{if } n < 0 \end{cases}$$

Note that the last notation here can be confusing when our group is an additive group, so we may consider the following alternative instead:

$$n \cdot a = \begin{cases} \underbrace{a + a + \cdots + a}_{n \text{ terms}} & \text{if } n > 0 \\ 0 & \text{if } n = 0 \\ \underbrace{a^{-1} + a^{-1} + \cdots + a^{-1}}_{n \text{ terms}} & \text{if } n < 0 \end{cases}$$

Theorem 1.1.14 ▶ Left and Right Cancellation Laws

Let G be a group, then for any $a, b, u, v \in G$,

- *if $au = av$, then $u = v$;*
- *if $ub = vb$, then $u = v$.*

Proof. Suppose that $au = av$, then

$$u = 1 \cdot u = \left(a^{-1}a\right)u = a^{-1}(au) = a^{-1}(av) = \left(a^{-1}a\right)v = 1 \cdot v = v.$$

The other case is similar. $\square$

A direct consequence of the cancellation laws is the following:

Corollary 1.1.15 ▸ Existence and Uniqueness of Solutions in Groups

Let G be a group, then for any $a, b, x, y \in G$, the equations $ax = b$ and $ya = b$ have unique solutions.

Proof. The uniqueness is obvious from Theorem 1.1.14, so it suffices to prove the existence. Take $x = a^{-1}b$, the obviously

$$ax = a\left(a^{-1}b\right) = \left(aa^{-1}\right)b = 1 \cdot b = b.$$

Therefore, $x = a^{-1}b$ is a solution to the equation $ax = b$. The other case is similar. □

A further consequence leads to the following way to identify the identity and inverses in a group:

Corollary 1.1.16 ▸ Characterisation of Identity and Inverses

Let G be a group. For any $a, b \in G$, if $ab = 1$ or $ba = 1$, then $b = a^{-1}$ and if $ab = a$ or $ba = a$, then $b = 1$.

Proof. Suppose that $ab = 1$, then by Corollary 1.1.15 we have $b = a^{-1} \cdot 1 = a^{-1}$. Similarly, suppose that $ab = a$, then we have $b = a^{-1}a = 1$. The other cases are similar. □

Lastly, note that we can take Cartesian product over groups.

Definition 1.1.17 ▸ Direct Product

Let $(A, \star)$ and $(B, \diamond)$ be groups. The **direct product** $A \times B$ is defined by the set

$$\{(a, b) : a \in A, b \in B\}$$

where

$$(a_1, b_1)(a_2, b_2) = (a_1 \star a_2, b_1 \diamond b_2).$$

One may check that for any groups A and B, the direct product $A \times B$ is always a group with identity $(1_A, 1_B)$ and inversion map $(a, b) \mapsto \left(a^{-1}, b^{-1}\right)$.

We will discuss the structure of matrix groups. However, as matrices are closely related to vector spaces and vector spaces are closely related to fields, it is helpful to first define what a field is.

Definition 1.1.18 ▸ Field

A **field** is a set $\mathbb{F}$ together with two binary operations $+$ and $\cdot$ such that

- $(\mathbb{F}, +)$ is an Abelian group with identity 0,
- $(\mathbb{F} \setminus \{0\}, \cdot)$ is an Abelian group with identity 1, and
- $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ for all $a, b, c \in \mathbb{F}$.

Remark. We use $\mathbb{F}^{\times}$ to denote the multiplication group $(\mathbb{F} \setminus \{0\}, \cdot)$.

The next proposition is an attempt in constructing a field of a fixed cardinality.

Proposition 1.1.19 ▸ Construction of a Finite Field

Let p be a prime number and $\sim$ be the equivalence relation on $\mathbb{Z}$ such that $a \sim b$ if and only if $a \equiv b \mod p$. Define $\mathbb{F}_p := \mathbb{Z}/\sim$ and

$$[x]_{\sim} + [y]_{\sim} = [x + y]_{\sim},$$
$$[x]_{\sim} \cdot [y]_{\sim} = [x \cdot y]_{\sim}$$

for all $[x]_{\sim}, [y]_{\sim} \in \mathbb{F}_p$, then $(\mathbb{F}_p, +, \cdot)$ is a field with $|\mathbb{F}_p| = p$.

Proof. Let $X := \{0, 1, \cdots, p - 1\}$ and define $f : X \to \mathbb{F}_p$ by $f(x) = [x]_{\sim}$. We claim that f is bijective. Suppose that $f(x) = f(y)$ for some $x, y \in X$, then $[x]_{\sim} = [y]_{\sim}$, which means $x \equiv y \mod p$. Without loss of generality, assume that $x \leq y$, then since $0 \leq y - x \leq p - 1$ we must have $y - x = 0$, so $x = y$. Therefore, f is injective. Take any $[x]_{\sim} \in \mathbb{F}_p$, then there exists some $b \in \mathbb{Z}$ and $r \in X$ such that $x = bp + r$, and so $x \equiv r \mod p$. This means that $\{x\}_{\sim} = [r]_{\sim} = f(r)$. Therefore, f is surjective and so is a bijection. Therefore, $|\mathbb{F}_p| = |X| = p$. □

Now we can fix some arbitrary field $\mathbb{F}$ and consider some matrices.

Definition 1.1.20 ▸ General Linear Group

Let $\mathbb{F}$ be a field, then for each $n \in \mathbb{Z}^+$, the **general linear group** of degree n on $\mathbb{F}$ is defined as

$$\mathrm{GL}_n(\mathbb{F}) := \{\boldsymbol{A} \in \mathcal{M}_{n \times n} : \det(\boldsymbol{A}) \neq 0\}$$

with respect to matrix multiplication.

Remark. In other words, the matrix group on $\mathbb{F}$ is the set of all non-singular $n \times n$ matrices on $\mathbb{F}$.

The reader might find it an interesting exercise to show that $\mathrm{GL}_n(\mathbb{F})$ is indeed a group. In particular, $\mathrm{GL}_1(\mathbb{F}) = \mathbb{F}^\times$.

1.2 Permutations

Next, we examine a more complicated example of groups known as *symmetric groups*. First, we shall prove a preliminary theorem.

Proposition 1.2.1 ▸ Bijectivity and Invertibility Are Equivalent

Let X, Y be any sets and let $\sigma \in Y^X$ be a map, then σ is invertible if and only if it is bijective.

Proof. Suppose that σ is invertible. Let $x_1, x_2 \in X$ be such that $\sigma(x_1) = \sigma(x_2)$, then

$$x_1 = \sigma^{-1}\big(\sigma(x_1)\big) = \sigma^{-1}\big(\sigma(x_2)\big) = x_2,$$

so σ is injective. Take some $y \in Y$, then there exists $\sigma^{-1}(y) \in X$ with $\sigma\big(\sigma^{-1}(y)\big) = y$, so σ is surjective. Therefore, σ is a bijection.

Conversely, suppose that σ is bijective. Define a map $\tau \colon Y \to X$ by $\tau(y) = x$ if and only if $\sigma(x) = y$. Since σ is injective, for every $y \in Y$ there is a unique x with $\sigma(x) = y$, so τ is well-defined. Take any $x \in X$ with $\sigma(x) = y$, we have

$$\tau\big(\sigma(x)\big) = \tau(y) = x,$$

so $\tau \circ \sigma = \mathrm{id}_X$. Similarly, $\sigma \circ \tau = \mathrm{id}_Y$. Therefore, σ is invertible. □

Now, the above proposition gives a way to quickly justify the existence of "inverses" for bijections, which we will use to prove the following result:

Proposition 1.2.2 ▸ Group of Bijections

Let Ω be a non-empty set and define $S_\Omega \subseteq \Omega^\Omega$ to be the set of all bijections from Ω to itself, then $(S_\Omega, \circ)$ is a group.

Proof. Trivial. □

Intuitively, if we label each element in Ω, the every map $\sigma \in S_\Omega$ is essentially a re-labelling of all elements via a one-to-one correspondence. Therefore, such bijections correspond exactly to the permutations of Ω.

Definition 1.2.3 ▸ Symmetric Group

For any non-empty set Ω, the set S_Ω of all bijective maps from Ω to itself is called the **symmetric group** under $\circ$. Each $\sigma \in S_\Omega$ is called a **permutation** of Ω.

Remark. In particular, for any $n \in \mathbb{Z}^+$, the group $S_n : S_{\{1,2,\cdots,n\}}$ is called the *symmetric group of degree n*.

Let us investigate a little bit about the cardinality of symmetric groups.

Proposition 1.2.4 ▸ Cardinality of Sets of Bijections

For any finite set Σ with $|\Sigma| = n$, define $S_{n,\Sigma}$ to be the set of all bijections from $\mathbb{Z}_n^+$ to Σ, where $\mathbb{Z}_n^+ = \{1, 2, \cdots, n\}$, then $|S_{n,\Sigma}| = n!$.

Proof. We shall proceed with induction on n. The case where $n = 1$ is obvious. Suppose that there exists some $k \in \mathbb{N}^+$ such that any finite set Σ_k with $|\Sigma_k| = k$ satisfies $\left|S_{k,\Sigma_k}\right| = k!$. Let Σ_{k+1} be any finite set such that $|\Sigma_{k+1}| = k + 1$ and consider

$$S_{k+1,\Sigma_{k+1}} = \bigsqcup_{a\in\Sigma} \left\{\sigma \in S_{k+1,\Sigma_{k+1}} : \sigma(k+1) = a\right\}.$$

It is clear that for each $a \in \Sigma_{k+1}$,

$$\left|\left\{\sigma \in S_{k+1,\Sigma_{k+1}} : \sigma(k+1) = a\right\}\right| = \left|S_{k,\Sigma_{k+1}\setminus\{a\}}\right| = k!.$$

Therefore,

$$\left|S_{k+1,\Sigma_{k+1}}\right| = |\Sigma_{k+1}| \cdot k! = (k+1)!.$$

□

By fixing $\Sigma = \{1, 2, \cdots, n\}$ in the above Proposition, we can easily deduce the following corollary:

Corollary 1.2.5 ▸ Cardinality of Symmetric Groups

For any $n \in \mathbb{N}^+$, we have $|S_n| = n!$.

Proof. Trivial. □

Note that this is exactly the same as the combinatorial argument that the total number of permutations over n elements is $n!$. Basically, to fix any bijection from Σ to $\mathbb{Z}_n^+$, we just need to take each element in Σ and "match" it to a unique unmatched element in $\mathbb{Z}_n^+$. In doing so, we are in fact applying the multiplication principle and it is not surprising that we can arrive at the same conclusion.

Notice that sometimes, if we apply a certain permutation repeatedly to a set, we might be able to restore the original arrangement of its elements. Such a situation is very peculiar and so we define the notion of *order* to describe it.

Definition 1.2.6 ▸ Order of a Group

Let G be a group. The **order** of $x \in G$, denoted by $|x|$, is the smallest $n \in \mathbb{N}^+$ such that $x^n = 1$. If for all $n \in \mathbb{N}^+$, we have $x^n \neq 1$, then we say that x has order ∞.

It is clear that all finite sets of cardinality n are equinumerous to S_n, so it suffices to study the behaviour of S_n's for all $n \in \mathbb{N}^+$. We first consider a special type of permutations which shift each element by one position.

Definition 1.2.7 ▸ Cycle

An **m-cycle** on S_n for some positive integer $m \leq n$ is the permutation over $\{a_1, a_2, \cdots, a_m\} \subseteq S_n$ such that

$$\sigma : a_i \mapsto \begin{cases} a_{i+1} & \text{if } 1 \leq i \leq m-1 \\ a_1 & \text{if } i = m \\ a_i & \text{otherwise} \end{cases},$$

denoted by $(a_1, a_2, \cdots, a_m)$.

To demonstrate that the cycle notation is a good notation, we offer the following proposition:

Proposition 1.2.8 ▸ Characterisation of Abelian Symmetric Groups

A symmetric group of degree n is not Abelian if and only if $n \geq 3$.

Proof. Let S_Ω be a non-Abelian symmetric group of degree n for some finite set Ω, then there exists $a, b \in S_\Omega$ such that $ab \neq ba$. Clearly, this means that $a \neq b$. Notice that a is not the identity because otherwise $ab = b = ba$, which is a contradiction. Similarly, b is not the identity, and so there must exists some $e \in S_\Omega$ which is the identity. Therefore, $n = |S_\Omega| \geq 3$.

Conversely, it suffices to prove that S_n is not Abelian if $n \geq 3$. Take $(1, 2), (1, 3) \in S_n$

and consider

$$(1,2)\circ(1,3)=(1,3,2)$$
$$(1,3)\circ(1,2)=(1,2,3)\,.$$

Clearly, $(1,3,2)\neq(1,2,3)$ and so $(1,2)$ and $(1,3)$ do not commute. Therefore, S_n is not Abelian. □

Remark. In fact, all non-Abelian groups must have cardinality at least 3.

Some rough observation shows that the reason that some cycles do not permute is that they contain overlapping elements which will get swapped to different positions if the order of the permutations is different. However, if two cycles are disjoint, then intuitively it should not matter which one is executed first.

Proposition 1.2.9 ▸ Commutativity of Cycles

Let $(a_1,a_2,\cdots,a_m),(b_1,b_2,\cdots,b_k)\in S_n$ be cycles, then

$$(a_1,a_2,\cdots,a_m)\circ(b_1,b_2,\cdots,b_k)=(b_1,b_2,\cdots,b_k)\circ(a_1,a_2,\cdots,a_m)$$

if $\{a_1,a_2,\cdots,a_m\}\cap\{b_1,b_2,\cdots,b_k\}=\varnothing$.

Proof. Define

$$\sigma_{ab}=(a_1,a_2,\cdots,a_m)\circ(b_1,b_2,\cdots,b_k)\,,$$
$$\sigma_{ba}=(b_1,b_2,\cdots,b_k)\circ(a_1,a_2,\cdots,a_m)\,,$$

then it is equivalent to proving that $\sigma_{ab}(x)=\sigma_{ba}(x)$ for all $x\in\mathbb{Z}_n^+$. Take any $x\in\mathbb{Z}_n^+$, if $x\notin\{a_1,a_2,\cdots,a_m\}\cup\{b_1,b_2,\cdots,b_k\}$, then $\sigma_{ab}(x)=x=\sigma_{ba}(x)$. If $x=a_i\in\{a_1,a_2,\cdots,a_m\}$, then

$$(a_1,a_2,\cdots,a_m)(x)=\begin{cases}a_{i+1} & \text{if } 1\leq i\leq m-1\\ a_1 & \text{if } i=m\end{cases},$$
$$(b_1,b_2,\cdots,b_k)(x)=x$$

because $\{a_1,a_2,\cdots,a_m\}\cap\{b_1,b_2,\cdots,b_k\}=\varnothing$. Therefore, $\sigma_{ab}(x)=\sigma_{ba}(x)$. The same argument applies to $x=b_j\in\{b_1,b_2,\cdots,b_k\}$. □

Those with the intuition or training in programming may realise that since a permutation of a set is essentially constructed by repeatedly swapping elements in pairs, we can naturally find a finite number of cycles in a permutation.

Definition 1.2.10 ▶ Cycle Decomposition

Let $\sigma \in S_n$ be any permutation, then a **cycle decomposition** of σ is a finite product in the form of

$$\sigma = \sigma_1 \circ \sigma_2 \circ \cdots \circ \sigma_k$$

where $\sigma_1, \sigma_2, \cdots, \sigma_k \in S_n$ are cycles.

Remark. Note that the cycle decomposition of a permutation is not unique in general.

Now, let us justify our programming intuition from a mathematical perspective.

Theorem 1.2.11 ▶ Existence of Cycle Decompositions

Every $\sigma \in S_n$ admits a cycle decomposition.

Proof. For each $\sigma \in S_n$, let us define D_σ to be a digraph with $V(D_\sigma) = \mathbb{Z}_n^+$ such that $x \to y \in E(D_\sigma)$ if and only if $\sigma(x) = y$. Fix any $v_0 \in V(D_\sigma)$. For every $i \in \mathbb{N}$, it is clear that $d^+(v_i) = 1$ and so we can take v_{i+1} as the unique out-neighbour of v_i. We claim that there exists some $v_k \in V(D_\sigma)$ such that $v_{k+1} = v_0$ because otherwise the graph is of an infinite order, which is not possible. Remove all edges $v_i \to v_{i+1}$ for $i = 1, 2, \cdots, k$ and let D'_σ be the resultant graph. We can repeat the same process again on D'_σ and we claim that eventually the graph will become empty. Suppose on contrary that the edge-minimal graph obtained through this algorithm is non-empty, then it must be acyclic and contains a longest directed path $u_0 u_1 \cdots u_k$. Let $\sigma(u_k) = w$, then $u_k \to w$ has been removed in a previous iteration, which means that $u_k \to w$ is contained in some cycle $\vec{C} \subseteq D$. Note that $u_0 u_1 \cdots u_k \nsubseteq \vec{C}$ because otherwise the path would have been removed, so there must exists a vertex $u_m \in u_0 u_1 \cdots u_k$ such that

$$u_m u_{m+1} \cdots u_k \subseteq \vec{C} \quad \text{but} \quad u_{m-1} \to u_m \notin E\left(\vec{C}\right).$$

However, this is not possible because this means that there exists some $u' \neq u_{m-1}$ such that $u', u_{m-1} \in N_D^-(u_m)$ and so $\sigma(u') = \sigma(u_{m-1}) = u_m$, which is a contradiction because σ is a bijection. Now, let $\vec{C_i}$ be the directed cycle removed at the i-th iteration, then by taking its vertices in clockwise order we form a cycle σ_i, where $i = 1, 2, \cdots, k$. It is clear that these cycles are pairwise disjoint and

$$\sigma = \sigma_1 \circ \sigma_2 \circ \cdots \circ \sigma_k.$$

□

The name "symmetric group" suggests a relation to geometry. We will now introduce a special group with some geometric interpretation.

Definition 1.2.12 ▸ Dihedral Group

For $n \in \mathbb{Z}_{\geq 3}$, the **dihedral group** of order $2n$, denoted by D_{2n}, is the group formed by $1, r, s$ such that

$$D_{2n} := \left\{1, r, r^2, \cdots, r^{n-1}, s, sr, sr^2, \cdots, sr^{n-1}\right\}$$

and

$$r^n = s^2 = 1 \quad \text{and} \quad rs = sr^{-1}.$$

The multiplication rule for a diheral group is a bit obscure, so we may wish to use the following technique to produce some readable notation:

Definition 1.2.13 ▸ Multiplication Table

Let $G := \{g_1, g_2, \cdots, g_n\}$ be a finite group. The **multiplication table** of G is an $n \times n$ matrix whose (i, j) entry is $g_i g_j$.

With "some" computation, one may notice that the multiplication tables of D_6 and S_3 look quite similar, which gives as the motivation to visualise D_{2n} using some geometric representation.

Consider a regular n-sided polygon P and define r as "clockwise rotation of P about its centre by $\frac{2\pi}{n}$" and s as "reflection of P about an axis through a fixed point in P and the origin". Surprisingly, such a definition is coherent with the multiplication rule in Definition 1.2.12! Indeed, doing r for n times and doing s twice both means that the polygon remains what it was originally, and a rotation followed by reflection is exactly the same as a reflection followed by a rotation in the opposite direction.

Remark. Note that for $n > 3$, the multiplication tables of D_{2n} and S_n no longer coincide because $|D_{2n}| = 2n \neq n! = |S_n|$.

1.3 Group Presentation

Definition 1.3.1 ▶ Generators

Let G be a group. A set of **generators** of G is a set $S \subseteq G$ such that every element of G can be written as a finite product of elements of S and their inverses.

We say that G is *generated* by S or S *generates* G, which is denoted by $G = \langle S \rangle$. We introduce some basic examples and conventions.

First, every group is generated by itself. In particular, every singleton group is generated by $\{1\}$ and $\varnothing$ (by convention). It is also clear that the addition group $(\mathbb{Z}, +)$ is generated by $\{1\}$ and that D_{2n} is generated by $\{r, s\}$. However, $(\mathbb{R}, +)$ is generated by $\mathbb{R}_{\geq 0}$ and not $\mathbb{Q}$ because irrationals cannot be written as a finite sum of rationals and their inverses. By Theorem 1.2.11, we also know that every symmetric group is generated by all of its cycles.

We also need to define how the elements in the generators should multiply with each other to properly "generate" the group. To do this, we make use of relations.

Definition 1.3.2 ▶ Relation in a Group

Let G be a group with generators S. A **relation** in G is a relation R on $S \cup \{1\}$.

Remark. Conventionally, it is useful to think of a relation in a group G as an equation to relate its generators to 1.

Roughly speaking, relations define how the generators can be combined together to derive all the other elements in a group. Therefore, we can fully specify a group using generators and relations.

Definition 1.3.3 ▶ Presentation

A **presentation** of a group $G = \langle S \mid R \rangle$ is a set of generators S and a set of relations R on S such that any other relation on S can be deduced from R.

1.4 Group Homomorphisms

Previously, we have seen how S_Ω and S_n are essentially "the same" if $|\Omega| = n$. This is exactly because of the fact that these groups have the same structure. We shall now formalise the notion of "having the same structure". In the context of groups, since every group can be completely defined by a binary operation, it is natural to define a structure-preserving transformation as one which preserves the binary operation.

Definition 1.4.1 ▸ Group Homomorphism

Let $(G, \star)$ and $(H, \diamond)$ be groups. A **homomorphism** from G to H is a map $\varphi : G \to H$ such that

$$\varphi(a \star b) = \varphi(a) \diamond \varphi(b)$$

for all $a, b \in G$.

Recall that the notions of identity and inverses are induced by the binary operation on a group, so naturally a group homomorphism should respect these structures as well.

Proposition 1.4.2 ▸ Properties of Group Homomorphisms

Let $\varphi : G \to H$ be a homomorphism, then

$$\varphi(1_G) = \varphi(1_H) \quad \text{and} \quad \varphi\left(a^{-1}\right) = \varphi(a)^{-1}$$

for all $a \in G$.

Proof. Notice that

$$\varphi(1_G)\varphi(1_G) = \varphi\left({1_G}^2\right) = 1_H\varphi(1_G),$$

so by 1.1.14, we have $\varphi(1_G) = 1_H$. Consider

$$\varphi\left(a^{-1}\right)\varphi(a) = \varphi\left(a^{-1}a\right) = \varphi(1_G) = 1_H.$$

Similarly, $\varphi(a)\varphi\left(a^{-1}\right) = 1_H$, so $\varphi\left(a^{-1}\right) = \varphi(a)^{-1}$ for all $a \in G$. □

Clearly, for any group G, the identity map $\mathrm{id}_G : G \to G$ is a homomorphism. Moreover, by denoting $\mathbb{1} := \{1\}$, the maps $\psi : \mathbb{1} \to G$ and $\varphi : G \to \{1\}$ with $\psi(1) = 1_G$ and $\varphi(a) = 1$ for all $a \in G$ are the unique homomorphisms for any group G. It is also clear from the above fact that "the" singleton group is unique up to homomorphisms.

Intuitively, since a homomorphism does not change the group's structure, the group's structure will not change after undergoing any sequence of homomorphisms.

Proposition 1.4.3 ▸ Composition of Homomorphisms Is a Homomorphism

For any groups G, H, K, if $\varphi : G \to H$ and $\psi : H \to K$ are homomorphisms, then

$$\psi \circ \varphi : G \to K$$

is a homomorphism.

Proof. Left to the reader as an exercise. □

Let us examine two simple groups, S_2 and S_3, and try to list down all homomorphisms between them.

Proposition 1.4.4 ▶ Homomorphisms from S_3 to S_2

Any group homomorphism from S_3 to S_2 is either the trivial identity map or the sign map $\operatorname{sgn}: S_3 \to S_2$ *such that*

$$\operatorname{sgn}(a) = \begin{cases} 1 & \textit{if } a = 1 \textit{ or } (123) \textit{ or } (132) \\ (12) & \textit{if } a = (12) \textit{ or } (23) \textit{ or } (13) \end{cases}.$$

Definition 1.4.5 ▶ Isomorphism

An **isomorphism** from G to H is a homomorphism $\varphi: G \to H$ such that there exists a homomorphism $\psi: H \to G$ such that $\psi \circ \varphi = \mathrm{id}_G$ and $\varphi \circ \psi = \mathrm{id}_H$

Proposition 1.4.6 ▶ Isomorphism Is Bijective

A homomorphism $\varphi: G \to H$ is an isomorphism if and only if it is bijective.